TELE9782 Technical Q&A

English answers for a 20–40 second exchange · 中文追问解释 · Q&A outside the main talk

For each answer, state the physical idea, give one project number, then identify the supporting slide. Source paths below are relative to D:/TELE9782/Project_Final.

01 Why did you choose a microstrip patch?

A single rectangular patch gives a low-profile geometry and a broadside beam. Its width and length can be estimated analytically, and the inset feed lets us tune the input resistance. The verified single element already exceeds the requirement.

结构简单、参数可解释。两辐射边形成前向主瓣，单元余量 0.96 dB，无需引入阵列馈电和互耦。

Slides 2, 3, 8 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv

02 Why D-band and why 150 GHz?

The assignment sets 130 to 170 GHz. We chose 150 GHz as the midpoint starting target and verified the final match and radiation there. The 40 GHz sweep describes the analysis range; the matched interval is 3.513 GHz wide.

先定任务频段，再定设计频点。带宽按 −10 dB 交点定义，不能拿 170−130 当匹配带宽。

Slides 2, 7 · Evidence: 02_Simulation_Results/S11.csv; Impedance_Bandwidth.csv

03 What is TM10 in this patch?

It denotes the dominant mode with one half-wave variation along the resonant length and approximately no variation across the width. The two open edges act as radiating slots whose fields add toward broadside.

本工程 L 沿 y，W 沿 x。模式下标跟随长度优先的记号，不能直接用下标猜全局坐标。

Slides 3, 16 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

04 How did you choose patch width?

The transmission-line starting estimate is c over twice the target frequency, multiplied by the square root of two over epsilon r plus one. With epsilon r of 3.5 at 150 GHz, this gives 0.666 millimetres. The final width is 0.667.

$W = c/(2f_0)\cdot\sqrt{[2/(\varepsilon r+1)]}$。W 同时影响辐射孔径、边缘电导及输入阻抗，公式给初值。

Slides 4, 11 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

Technical Q&A · 05–08

05 How did you choose patch length?

I estimate the effective half-wave length with the effective permittivity, then subtract the fringing extension at both edges. The starting physical length is 0.520 millimetres, and full-wave tuning gives 0.513 millimetres.

$L_{eff}=c/(2f_0\sqrt{\varepsilon_{eff}})\approx 0.5584$ mm，$\Delta L\approx 0.0190$ mm，$L=L_{eff}-2\Delta L\approx 0.5204$ mm。

Slides 4, 11 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

06 Why use effective permittivity?

The microstrip fields occupy both the substrate and the surrounding air. The effective permittivity approximates that mixed environment for phase propagation. We use about 3.203 to estimate the guided wavelength, compared with 3.5 for the bulk substrate.

ε_{eff} 处于空气与介质之间。不要把它当成另一个材料属性填进 CST；CST 使用基板 $\varepsilon_r=3.5$。

Slides 3, 11 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

07 Why do fringing fields matter?

Electric fields extend beyond the metal at the open edges. They make the patch electrically longer than its physical length and provide the mechanism for radiation. The length correction accounts for both edges.

边缘场同时解释 ΔL 和辐射缝隙模型。若直接把 0.558 mm 当物理 L，通常会把谐振估低。

Slides 3, 11 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

08 Why is the length about half a guided wavelength?

The fundamental standing wave has one half-cycle along the effective length, corresponding to a phase change of about pi. The finite open edges introduce fringing, so this condition applies approximately to effective length.

$\beta L_{eff}\approx\pi$；$L_{eff}=L+2\Delta L$；$\beta\approx 2\pi\sqrt{\varepsilon_{eff}}/\lambda_0$。这里讨论基模的解析近似。

Slides 3, 11 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

Technical Q&A · 09–12

09 What determines patch directivity?

It depends on the angular distribution of radiated power. The two radiating edges, their phase relationship and the finite ground shape that distribution. Our full-wave calculation includes the complete fed structure, including its edge fields.

Dmax=4πUmax/Prad。孔径及场相位决定峰值与全空间平均的比。单靠尺寸公式不能直接得 7.460。

Slides 3, 8, 16 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv

10 Why did you use inset feeding?

It lets the microstrip contact the patch at a point with a more suitable input resistance. The feed stays planar and needs no external transformer section. The sweep identifies a practical depth for this geometry.

边缘电压高、等效输入电阻高。向内移可以改变驱动点阻抗；间隙与馈线同时产生局部不连续。

Slides 6, 13 · Evidence: 05_Documentation/Iteration_Log.csv

11 Why does the inset change input resistance?

The voltage and current change along the standing wave. Moving the contact from the edge towards the centre lowers the resistance in the simple model. I used that trend to guide the sweep and then checked the full-wave reflection response.

$R_{in}(d) \approx R_{edge} \cos^2(\pi d/L)$，d 从馈电辐射边算。实际结果还含 inset 缝隙耦合和输入电抗。

Slides 6, 13 · Evidence: 05_Documentation/Iteration_Log.csv

12 Why was the 0.190 mm inset selected?

In the early controlled trials, 0.17 millimetres gave a minimum near minus 14.18 dB and 0.19 gave minus 26.82 dB. We kept 0.19 and shortened the patch to move the matched region towards 150 GHz.

两个 S_{11} 最低值发生在各自的频率，不能理解为 150 GHz 的逐点改善。早期远场属于诊断。

Slides 6, 13 · Evidence: 05_Documentation/Iteration_Log.csv

Technical Q&A · 13–16

13 What is the purpose of the waveguide port?

It solves the transverse microstrip mode and launches it into the model. The port crosses the feed, dielectric and ground reference at the end of a uniform section, with the propagation direction towards the patch.

端口是横截面模态边界。CST 的 ymin 在此表示向 +y 激励。端口参考面位移设 0。

Slides 5, 13 · Evidence: 04_Automation/engineering.py; 02_Simulation_Results/CST_Raw/1D Results_Port Information_Line Impedance_1(1).csv

14 Why choose 50 ohms and what does 49.47 mean?

Fifty ohms is the reference used for reporting S parameters. The final port eigenmode line impedance is 49.47 ohms at 150 GHz, which is close to that reference. The earlier baseline line impedance was approximately 49.9 ohms.

区分 S 参数归一参考、馈线模态特性阻抗和天线输入阻抗。三者都可用 Ω 表示，物理定义不同。

Slides 5, 13 · Evidence: 04_Automation/engineering.py; 02_Simulation_Results/CST_Raw/1D Results_Port Information_Line Impedance_1(1).csv

15 What does S11 of minus 13.424 dB mean?

It is twenty log ten of the reflection-coefficient magnitude. Squaring the linear magnitude gives a reflected power fraction of about 4.55 percent. The remaining 95.45 percent is accepted at the port.

$|S_{11}|=10^{(-13.424/20)}\approx 0.213$；反射功率比例约 0.04545。接受功率还要扣除金属和介质耗散。

Slides 7, 14 · Evidence: 02_Simulation_Results/S11.csv; Impedance_Bandwidth.csv

16 Why is the deepest dip at 148.84 GHz?

The S11 dip is the best input match on the sample grid. The finite ground, feed geometry and reference impedance influence its location. The design frequency of 150 GHz is still inside the matched interval and has its own farfield monitor.

最低反射频率不等于已证明的孤立本征谐振。解析最终尺寸给 152.000 GHz，比较误差 2.08%。

Slides 4, 7 · Evidence: 02_Simulation_Results/S11.csv; Impedance_Bandwidth.csv

Technical Q&A · 17–20

17 How did you achieve greater than 6.5 dBi?

The single grounded patch supports a broadside mode. We chose dimensions near the desired frequency, matched the input and verified the full-wave farfield at 150 GHz. Its peak is 7.460 dBi, which is 0.960 dB above the target.

物理原因是前向辐射集中。匹配使工作点可用，正确端口和远场设置使数字可信。

Slides 3, 8, 10 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv

18 What is the difference between directivity and gain?

Directivity compares directional intensity to the average over the sphere using total radiated power. Gain includes radiation efficiency, so it also accounts for accepted power dissipated as heat. The angular shape may be similar while the level changes.

$D=4\pi U/P_{rad}$，$G=4\pi U/P_{acc}=\eta_{rad} D$。全部线性表示；转换成 dBi 用 $10\log_{10}$。

Slides 8, 9, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

19 How does realized gain differ from gain?

Realized gain also includes the mismatch factor at the reference impedance. A port may reflect part of the incident power even when the accepted power is efficiently radiated. At 150 GHz, mismatch multiplies gain by about 0.9545.

$G_{real}=G(1-|S_{11}|^2)$。分母依次是 Prad、Pacc、Pincident。

Slides 8, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

20 Why is realized gain only 5.804 dBi?

At 150 GHz, radiation efficiency is 0.7155 and the accepted fraction is 0.9545. Their product is 0.6830. In decibels, adding ten log ten of 0.6830 to 7.460 gives 5.804. The criterion uses directivity.

方向性和实现增益相差约 1.656 dB。功率链解释差值，两个结果可以同时成立。

Slides 8, 9, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

Technical Q&A · 21–24

21 What is radiation efficiency?

Radiation efficiency is radiated power divided by accepted power. In this model the corresponding powers at 150 GHz are about 0.34150 and 0.47727 watts under CST's excitation normalization, giving 71.55 percent.

ηrad=Prad/Pacc。总效率分母是入射功率，0.34150/0.5≈68.30%。0.5 W 是仿真归一化，不是实物输入。

Slides 9, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

22 How was the minus-ten-dB bandwidth calculated?

I interpolated between the S11 samples on either side of each minus-ten-decibel crossing. The bounds are 147.106 and 150.619 GHz. Subtracting gives 3.513 GHz, and dividing by the midpoint gives 2.36 percent.

BW=fH−fL；FBW=BW/[(fH+fL)/2]×100%。交点是线性插值，谷底取实际扫频网格点。

Slides 7 · Evidence: 02_Simulation_Results/S11.csv; Impedance_Bandwidth.csv

23 Why does it not match over 130 to 170 GHz?

The patch is a resonant element, and its input impedance changes with frequency. The coursework specifies a design range and a directivity target. This design provides useful matching over a 3.513 GHz interval around the selected point.

窄带响应与储能、辐射和匹配有关。扩带需要单独设计验证，不能把整个扫频都报成工作带宽。

Slides 2, 7 · Evidence: 02_Simulation_Results/S11.csv; Impedance_Bandwidth.csv

24 Why a single patch rather than an array?

The single patch already reaches 7.460 dBi at the matched point. An array could narrow the beam, but it would introduce a distribution network, additional losses and mutual coupling. A single element meets this project objective with fewer variables.

本项目阵列仅是可选方案。需要对相位、幅度、间距、馈电损耗与匹配分别验证。

Slides 2, 8 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv

Technical Q&A · 25–28

25 Why was the old maximum negative?

For a normalized full sphere the mean linear directivity is one, so its maximum cannot be below one, or zero dBi. The old port orientation and power inconsistency made its reported maximum unreliable. A negative value in an individual direction is possible.

D 的球面平均为 1。历史 −20.92 是最大值异常，不是物理天线从负值升到正值的证据。

Slides 6, 17 · Evidence: 05_Documentation/Project_Diagnosis.md; Simulation_Validation.md

26 How did you identify the port problem?

The original port used ymax and fed toward negative y, away from the patch. We checked the history and propagation convention. Reversing it exposed a short uniform section, which is why we reconstructed the launch.

看 history 中 Orientation，再看端口截面与馈线位置。改变符号后的失败与均匀段不足是两个具体问题。

Slides 5, 6, 13 · Evidence: 05_Documentation/Project_Diagnosis.md; Simulation_Validation.md

27 Why is a uniform feed section needed?

The port solver assumes a well-defined transverse mode at its reference section. A uniform launch separates the port from nearby discontinuities and meets CST's requirement for homogeneous mesh planes along propagation.

不能让缝隙或扩展地突变直接落在端口附近。均匀段帮助区分馈线本征模与不连续局部场。

Slides 5, 13 · Evidence: 04_Automation/engineering.py; 02_Simulation_Results/CST_Raw/1D Results_Port Information_Line Impedance_1(1).csv

28 What is the farfield integration surface?

It encloses the sources and uses the computed fields to evaluate radiation at large distance. The near-model option gave an inconsistent normalization in this model. The accepted run uses the global enclosing surface and a stricter energy-decay limit.

等效面近远场变换需要可靠的切向场和功率归一。近模型测试球均值 1.291 被拒；最终 0.9991。

Slides 6, 14, 17 · Evidence: 05_Documentation/Project_Diagnosis.md; Simulation_Validation.md

Technical Q&A · 29–32

29 Why use open boundaries?

They approximate an unbounded region so outgoing waves can leave with low reflection. The model uses added space on the non-feed faces and an open feed-side face with the physical port. This is appropriate for the radiating antenna.

近场与吸收边界需要间距。PML 间距参考为 130 GHz；端口是在 ymin 实际馈线截面。

Slides 5 · Evidence: 04_Automation/engineering.py; 02_Simulation_Results/CST_Raw/1D Results_Port Information_Line Impedance_1(1).csv

30 What does mesh convergence mean here?

We compare the same geometry and farfield setup at two mesh resolutions. The 150 GHz directivity moves by 0.037 dB and the match frequency by 0.56 percent. That supports the operating-point conclusion.

控制变量：同几何、同边界、同远场面、同终止条件。76,860 和 262,197 cells 不是两个不同天线。

Slides 10, 14 · Evidence: 02_Simulation_Results/Mesh_Convergence.csv

31 Does the mesh comparison prove full convergence?

It gives a sensitivity check for the quantities we compared. The depth of the sharp S11 dip and off-band details can move more. A third resolution or a second solver would help bound those uncertainties.

两点差小不能证明渐近收敛或给严格误差界。答辩只主张 150 GHz D 与匹配谷频率在这两档变化小。

Slides 10, 14 · Evidence: 02_Simulation_Results/Mesh_Convergence.csv

32 How did you check numerical reliability?

We checked passivity, full-sphere normalization, power balance and mesh sensitivity. The accepted pattern has mean linear directivity of 0.9991. The delivered project also reopened and reran successfully, with very small differences in exported directivity.

这是独立于看曲线外形的数值复核。与物理测试相比仍同源于 CST，材料偏差无法由复跑排除。

Slides 10, 14 · Evidence: 02_Simulation_Results/Verification.json; Sphere_Normalization_Check.json; 05_Documentation/Simulation_Validation.md

Technical Q&A · 33–36

33 How was the power-balance residual calculated?

We subtract radiated power, metal loss and dielectric loss from accepted power. At 150 GHz the remaining difference is about 0.00360 watts. Dividing by accepted power gives 0.754 percent.

Pacc−Prad−Pmetal−Pdiel≈0.003596 W；除以 0.477274 W。反射已从 Pacc 中扣除。

Slides 10, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

34 Are the material properties accurate at 150 GHz?

The model uses nominal polyimide parameters of 3.5 and 0.0027 and finite conductivity for gold. They are documented assumptions. Material characterization would be needed to quantify dispersion and actual sample losses at D-band.

介电常数偏差会改变电长度，损耗角正切会改变效率。仿真数据没有验证实际材料参数。

Slides 4, 12 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

35 What are the main model limitations?

The material values are nominal, conductor roughness and fabrication tolerance were not characterized, and the mesh test covers only two resolutions. The port eigenmode approximation is also simpler than the lossy full three-dimensional model.

区分端口求模近似与 3D 材料损耗。实际基板批次、尺寸误差和表面粗糙度都可能改变结果。

Slides 5, 10, 12 · Evidence: 02_Simulation_Results/Verification.json; Sphere_Normalization_Check.json; 05_Documentation/Simulation_Validation.md

36 Why was gold selected?

The original engineering model already specified gold and polyimide. We retained those materials and used finite gold conductivity, 4.561 times ten to the seventh siemens per metre. This keeps the loss calculation consistent with the project assumptions.

这是沿用工程材料定义，不主张金是所有候选中最优。0.002 mm 金属有有限导电损耗。

Slides 4, 12 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

Technical Q&A · 37–40

37 How does conductor loss affect the results?

Some accepted power is dissipated in the metal rather than radiated. That lowers radiation efficiency and gain. The effect on directivity depends on whether the loss also changes the angular field distribution.

150 GHz 金属损耗约 0.08954 W，介质损耗约 0.04264 W。D 的分母是 Prad，不能简单按损耗比例减 D。

Slides 9, 14 · Evidence: 02_Simulation_Results/Power_Consistency.csv; Radiation_Efficiency.csv; Realized_Gain.csv

38 What if patch length or substrate thickness changes?

Shortening the patch generally raises its resonant frequency. Changing thickness affects effective permittivity, fringing, line impedance and bandwidth together. I would recalculate the starting geometry and re-tune the feed before making a quantitative claim.

f≈c/[2(L+2ΔL)√εeff]。h 同时影响 W/h、ΔL 和 feed 的 w/h。单参数趋势有条件。

Slides 4, 11, 13 · Evidence: 05_Documentation/Engineering_Design_Notes.md; Analytical_Starting_Design.json

39 Could an array improve directivity?

An array can concentrate radiation by combining element fields in phase toward the desired direction. Its performance would depend on element spacing, coupling and the feed network. The single element already meets our requirement at 150 GHz.

阵因子乘以单元方向图是初步模型；实际全波阵列要考虑互耦、寄生波瓣、功分与损耗。

Slides 2, 8 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv

40 Are these experimental measurements?

They are numerical results from a full-wave CST model. The transmission-line equations gave starting dimensions, and CST predicted the response of the detailed model. Measurements would require fabricating and testing a prototype with a calibrated D-band setup.

三层证据：理论初算 → 数值仿真验证 → 实物实验。本项目完成前两层，未测实物。

Slides 4, 10 · Evidence: 02_Simulation_Results/Verification.json; Sphere_Normalization_Check.json; 05_Documentation/Simulation_Validation.md

Technical Q&A · 41–42

41 How did you label E- and H-plane?

At broadside the exported electric field is mainly along y. The plane containing that field and the broadside z direction is yz, the E-plane. The orthogonal xz plane is the H-plane.

yz 为 ϕ=90°/270°，xz 为 ϕ=0°/180°。极坐标切面展示总方向性，标签来自实际极化。

Slides 9, 16 · Evidence: 02_Simulation_Results/Farfield_150_Sphere.txt; E_Plane.csv; H_Plane.csv

42 Why not claim the peak at 154 GHz?

154 GHz has the highest sampled directivity, 7.570 dBi, but S11 there is only about minus 3.44 dB. We report 150 GHz because it has useful matching and clears the directivity threshold at the same frequency.

154 GHz 仅是监视点方向性最大；149 GHz 实现增益最大 5.982 dBi。报告三种最值时要带上各自频率。

Slides 7, 8, 15 · Evidence: 02_Simulation_Results/Directivity.csv; Validation_Summary.csv